LA GRANDEE INTERNATIONAL COLLEGE

Simalchaur, Pokhara Nepal

A Project Proposal

On

"Rock, paper, scissor "

Submitted to:

Bachelor of Computer Application(BCA) Program

In partial fulfilment of the requirements for the degree of BCA under

Pokhara University

Submitted by:

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Acknowledgement

The satisfaction that accompanies after the successful completion of any task will be incomplete without mentioning the people whose ceaseless and relentless cooperation, constant guidance and encouragement made this project possible.

We are grateful to our project supervisor **Mr Arpan Pokhrel**, our faculty teacher **Mr. Ramesh Chalise,** for the guidance, inspiration and constructive suggestions that helped us in the preparation of this project.

We are also appreciative among each other and have understood that teamwork, the designation of the task per the skillset one portrays, constant synchronisation and monitoring of progress and instilling new knowledge and skill is imperative for the success of any given work.

Sincerely,

Divya Sharma

Sabina Puwanli

Declaration for

“Project Title”

Student’s Declaration

We hereby declare that we are the only authors of this work and that no sources other than the listed here have been used in this work.

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Supervisor's Declaration

I hereby recommend that this project entitled "**Rock, Paper, Scissor**" is done under my supervision by **Sabina Puwanli and Divya Sharma** during their 2nd Semester in partial fulfilment of the requirements for the degree of **Bachelor in Computer Application** under **Pokhara University** is completed to my satisfaction and be processed for final evaluation.

____

Name of the Supervisor

Date:____ / ____ / _________

Letter of Approval

We certify that we have examined this report entitled "**Rock, Paper, Scissor**" and are satisfied with the project defence. In our opinion it is satisfactory in the scope and qualify as project in partial fulfilment of the requirements for the degree of **Program Name** under **Pokhara University.**

___________________________	___________________________	___________________________
Supervisor	**Examiner**	**Program Coordinator**
Arpan Pokhrel		

Date:____ / ____ / ________

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ABBREVIATIONS

A/D	Analog-to-Digital
ABC	Atanasoff Berry Computer
ACM	Association for Computing Machinery
AI	Artificial Intelligence
ALGOL	Algorithmic Language
ALU	Arithmetic Logic Unit
AMD	Advanced Micro Devices
ARPANET	Advanced Research Project Agency Network
ASCII	American Standard Code for Information Interchange
BASIC	Beginners All-purpose Symbolic Instruction Code
BCD	Binary Coded Decimal
BIOS	Basic Input Output System
BIPS	Billions of Instructions Per Second
BPI	Bytes Per Inch
CAD	Computer Aided Design
CAE	Computer Aided Engineering
CAN	Campus Area Network
E.G.	Example

1. Introduction

Rock, Paper, Scissors is one of the most popular and oldest hand games played throughout the world. The game is simple, entertaining and commonly used for decision making purposes. Traditionally , the game is played between two players where each player chooses wheather rock, paper, or scissors .In this computerized version, the user plays against the computer ,and the winner is determined according to the rules: rock defeats scissors, scissors defeat paper, and paper defeats rocks.

This project focuses on developing an advanced Rock Paper Scissors game using C programming language with several advanced features such as login system, registration system, CRUD operations, score management, multiplayer gameplay, account deletion, tournament mode, leaderboard system, match history, and file handling .The project is designed to simulate a real-world gaming system where users can create accounts, store scores permanently, compete with other players, and manage their data efficiently .

The system allows users to create accounts and securely log in using usernames and passwords. User data is stored permanently using file handling techniques .Once logged in, players can play against the computer or compete with another player in multiplayer mode. Scores are recorded and stored in files so that player performance can be tracked even after the program is closed.

The implementation of CRUD operation is one of the most important highlights of the project. CRUD stands for Create, Read, Update, and Delete. In this project, users can create new accounts, read player details and scores, update records automatically through gameplay, and delete accounts permanently .These operations are commonly used in real world database systems, making the project highly practical and educational.

File handling is another important component of the project .Since the C language does not include a built-in database management system, text files are used to store player records, scores, and match history. The multiplayer feature increases interaction and competitiveness between users. Instead of limiting gameplay to computer-based interaction only, the system allows two users to compete directly against each other. This improves our experience and makes the game more engaging.

This project also includes additional innovative feature such as tournament mode and streak bonus system. Tournament mode allows users to compete in multiple rounds continuously, while the streak bonus system reward users for consecutive wins. These features increase excitement and provide better gaming experience.

The project demonstrates the practical application of various programming concepts including loops, conditional statements, arrays, structures, functions, file handling and user authentication which helps beginner programmers understand how multiple concepts can be combined together to create a complete software application.

Overall, the advanced Rock Paper Scissors game is an educational and interactive project that combines entertainment with software development knowledge. It provides practical understanding of game logic, user management systems, and permanent data storage through file handling.

Objective

The main objective of this project is to develop a complete Rock Paper Scissors gaming system using C programming language. Here are some specific objectives listed below:

1. To develop an interactive Rock paper Scissors game using C programming with user-friendly gameplay features.
2. To implement multiple gameplay modes including player vs computer and multiplayer functionality.
3. To use file handling for storing player records, maintaining match history, and displaying scoreboards.
4. To enhance logical thinking and practical programming skills through the implementation of game logic and CRUD operations.

Study of Previous System

The previous Rock Paper Scissors system was a simple game developed to provide basic entertainment for users. The system allowed a player to play against the computer by selecting one of three options: rock, paper, or scissors. The computer generated its choice randomly, and the winner was decided according to the game rules. The system was easy to use and helped user understand the basic concept of game programming.

Features of the Previous System

1. Basic single-player gameplay.
2. Random computer-generated moves.
3. Simple game logic implementation.
4. Basic score display during gameplay.
5. Simple user interface.

Limitations of Previous System

1. The game was limited to single-player mode only.
2. There was no user registration and login system.
3. Scores and match history were not stored permanently.
4. No multiplayer game feature was available.
5. The game lacked a scoreboard and ranking system.
6. CRUD operations were not implemented properly.
7. The interface was less interactive and user-friendly.
8. Data handling and file management were inefficient.
9. The game had limited features and low user engagement.

Current Problem and Proposed Solution

Current Problems

1. There Traditional games do not save player records permanently.
2. Users cannot create personal accounts.
3. is no secure login system.
4. Players cannot track scores and ranking.
5. Multiplayer mode is missing in basic systems.
6. Existing systems lack CRUD operations.
7. Match history cannot be stored.
8. Most systems are not educational for programming students.

Proposed Solutions

To solve the above problems, an advanced Rock Paper Scissors game will be developed using C programming language.

The proposed system will include:

1. User registration and login system.
2. File handling for permanent data storage.
3. Multiplayer gameplay mode.
4. Score management system.
5. Match history feature.
6. Leaderboard system.
7. CRUD operations.
8. Tournament mode.
9. Streak bonus feature.

The proposed system will improve user interaction, provide better learning opportunities, and demonstrate real-world software development concepts.

High Level Requirements

High level requirements describe the major features and functionalities that the proposed Rock Paper Scissors game system should provide. These requirements define what the system is expected to achieve without going into technical details.

Functional Requirements:

Functional requirements describes the functions and services provided by the system.

1. The system should allow users to register accounts.
2. The system should allow users to log in.
3. The system should verify usernames and passwords.
4. The system should allow players to play both single-player and multiplayer modes.
5. The system should maintain scores.
6. The system should display leaderboard rankings.
7. The system should save match history.
8. The system should allow account deletion.
9. The system should store data using file handling.
10. The system should support tournament mode.
11. The system should provide streak bonus rewards.

Non-Functional Requirements:

Non-functional requirement describe the quality and performance of the system.

1. The system should be user-friendly.
2. The system should execute efficiently.
3. The system should store data securely.
4. The system should be easy to maintain.
5. The system should provide fast response time.
6. The system should be reliable.
7. The system should handle file operations properly.
8. The system should be simple for beginners to use.
9. The system should provide accurate results.

Requirement Prioritization Table

S.N	Requirement	Priority Level	Description
1	User registration and login	High	Required for identifying and managing users.
2	Single-Player mode	High	Core gameplay feature against computer
3	Multiplayer mode	High	Enhances user interaction and engagement
4	Player choice selection (rock/paper/scissors)	High	Basic game functionality
5	Winner determination	High	Core game logic
6	Score display	High	Shows result after each round
7	File handling (store scores and history)	High	Ensures permanent data storage
8	CRUD operations	High	Required for managing user data
9	Scoreboard and ranking system	Medium	Improves competitiveness
10	View match history	Medium	Allows users to track past games
11	Logout and exit system	low	Secondary system control feature
12	User-friendly interface	High	Improves usability and experience

System Design

System design describes the overall structure and working mechanism of the Rock Paper Scissor game .It explains how different modules of the system interact with each other, how data is processed, and how the user interact with the application.

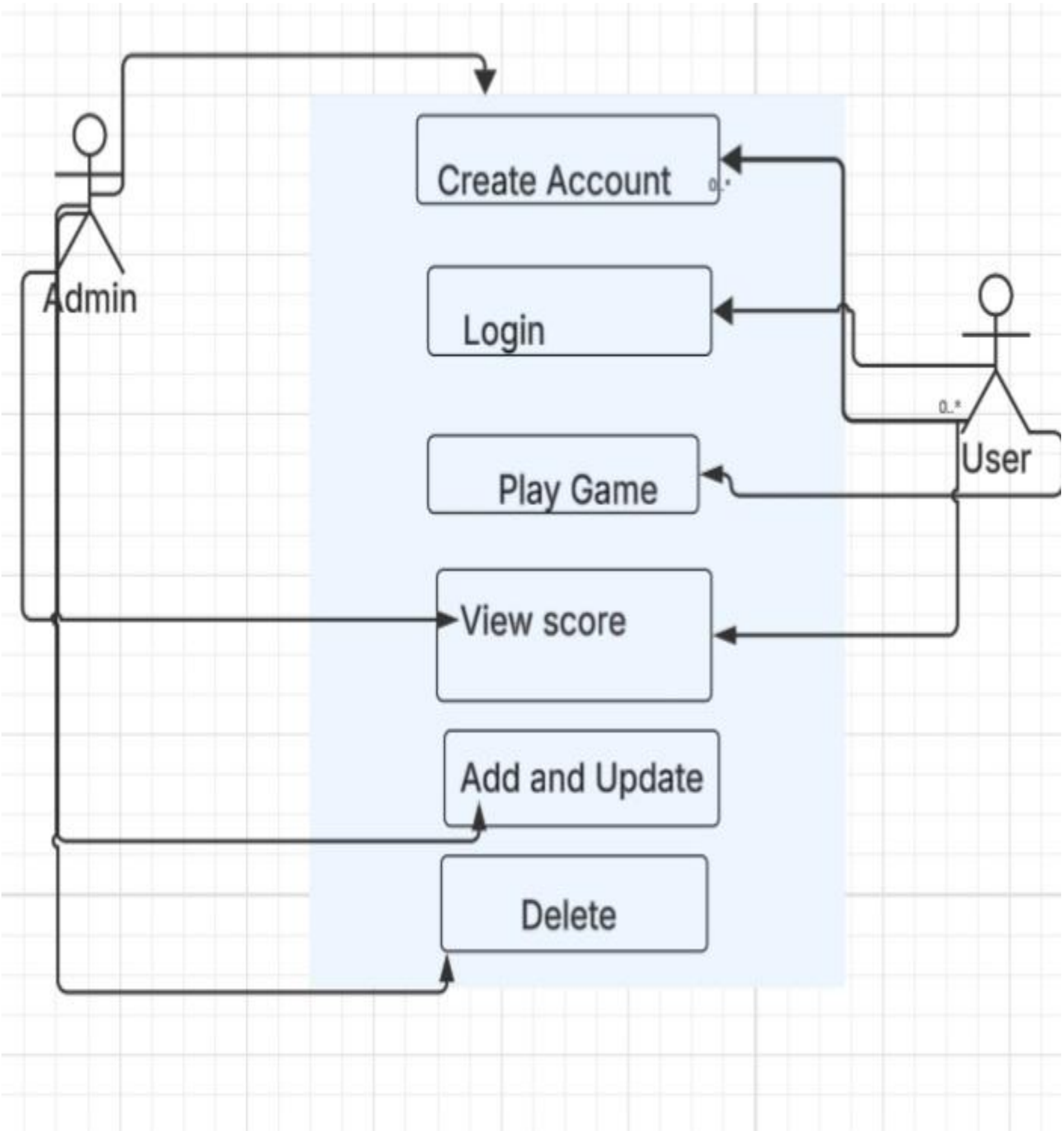

Figure 1Use Case diagram

Detailed Timeline

Introduction

The project was developed using the Waterfall model. The Waterfall model was selected for the development of Rock Paper Scissors game because it follows a linear and sequential approach. This model is suitable for small-scale projects where requirements are clearly defined at the beginning and are not expected to change frequently.

Since the system requirements such as user authentication, gameplay logic, score management, and match history were already identified, the waterfall model provided a clear structure for development. It also supports proper documentation at each phase, which is essential for academic projects.

Phases of Waterfall model Applied

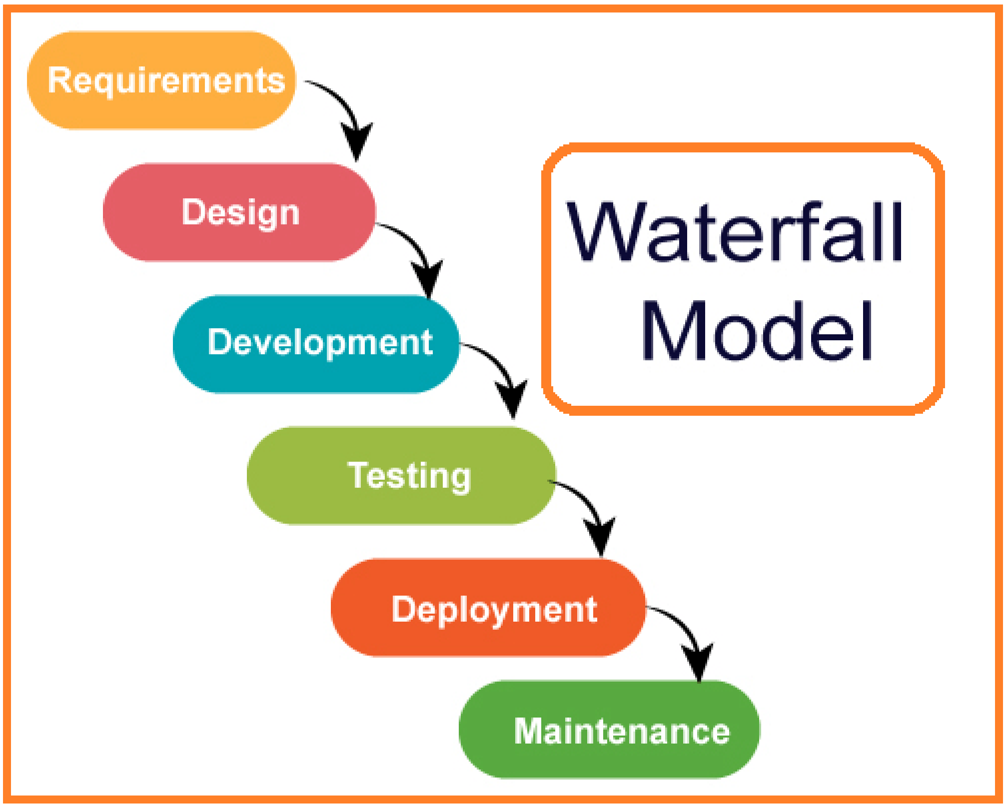

Figure 2waterfall model